Melissa Regalado

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EDUCATION

Boston University College of Engineering — Boston, MA
Bachelor of Science in Computer Engineering — May 2026

SELECTED PROJECTS

Low-Power HBLSA SRAM Design Simulation — May 2025

- Designed and simulated a 6T Static Random Access Memory (SRAM) cell, a six-transistor memory circuit used for fast on-chip data storage, implementing a Hierarchical Bit-Line with Local Sense Amplifiers (HBLSA) to reduce dynamic power.
- Assisted with transistor sizing, schematic development, and pre-recharge/sense amplifier circuit simulation to tune design parameters, validate results, and compare performance metrics.

Real-Time Image Processing System — Dec 2024

- Directed system architecture design, resolved UART timing issues, and guided integration of the MATLAB-FPGA workflow for seamless image transfer and display.
- Led development of a hardware-accelerated image processing pipeline on a heterogeneous system integrating MATLAB with an FPGA, a reconfigurable platform for fast computation and real-time digital processing, enabling image darkening, brightening, and inversion via UART.

Servo-Actuated Assistive Eating Utensil — May 2024

- Designed and prototyped a mechanically stabilized utensil for users with tremors or limited dexterity, achieving 100% pitch compensation and 76% roll stabilization through dual-axis servo control with motion filtering.
- Led circuit design, embedded programming, and testing trials, enabling reliable gesture differentiation and driving iterative improvements to control logic and system performance.

PROFESSIONAL EXPERIENCE

Program Coordinator — Jul – Sept 2025
Touch Education Technology — Cambridge, MA

- Enabled the successful delivery of AI+X-integrated STEM programming by managing logistics, supporting instruction, and fostering cross-disciplinary learning environments.
- Ensured student safety and engagement during Boston-based workshops through hands-on supervision and local guidance.
- Maintained clear communication between students, mentors, and leadership to foster program cohesion and community.

Senior Student Admissions Representative (Sr. SAR) — May 2023 – Present
Boston University Undergraduate Visitor Center — Boston, MA

- Coordinate and schedule ~100 current BU students for 200+ prospective student conversations each semester, strengthening engagement through Virtual Student Chats.
- Delivered 50+ campus tours as a BU Tour Guide and Scarlet Speaker, representing BU's mission and fostering inclusive experiences.
- Selected to the Sr. SAR leadership team (Fall 2025) to oversee Virtual Events & Programs and support peer representatives.

Tutor Mentor at Boston University — Jun – Aug 2025
Upward Bound Math & Science — Boston, MA

- Delivered academic mentorship in math and science, enhancing student performance and readiness for college-level STEM coursework.
- Empowered first-generation and underrepresented students by cultivating confidence, study strategies, and long-term academic success.

TECHNICAL SKILLS

Programming: Python, C, C++, Java, Verilog, Assembly, MATLAB, HTML/CSS, Bash
Tools & Hardware: Cadence Virtuoso, Arduino, FPGA, Git, Unix/Linux, Soldering, Oscilloscope, Drill Press
Languages: Spanish (Fluent)

CAMPUS INVOLVEMENT

Dean's Host & Tour Guide – College of Engineering — Fall 2025 – Present
Member – BU Admissions Student Diversity Board — Fall 2023 – Present
Secretary – First-Generation Low-Income Partnership (FLIP) Executive Board — Fall 2022 – Spring 2023